SAT Resetmaxxing for Fun and Profit

(Final Title TBD)

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Abstract

We improve SATCoP performance by resetting the connection tableau often. This causes more and more varied SAT statements to be added to the SAT prover, which causes it to saturate faster.

1 Introduction

2 Preliminaries

(TODO: Tableau or Tableaux? Bibliography is inconsistent.)

2.1 Tableau Solving Methods

Minimalist tableau-style provers based on Prolog have been used with great success since the release of leanTAP in 1995[1], and have consistently achieved high rankings in solving FOF and CNF problems in the Thousands of Problems for Theorem Provers (TPTP) library[7].

Rather than working like saturation provers (TODO: what do saturation provers even *do*?) such as Vampire prover[5], Tableau-based provers attempt to create a Connection Tableau with ground terms. The prover builds a tree, where each node corresponds to a particular clause of the problem in conjunctive normal form, using one of the following four steps.

(TODO: Add a diagram and/or a description.)

Figure 1: Description of the operations of Tableau calculus[2].

LeanCoP[4] improves
Talk about LeanCoP[3], possibly MeanCoP[2].

2.2 SATCoP and Similar Methods

3 Research Vehicle

Talk about the Connections Prover[6].

4 Experimental Evaluation

5 Future Results

References

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